Universal manuscript template for Optica Publishing Group journals

YASH PATEL,[1] PRAJAKTA BELEKAR,[1] GEORG LADURNER,[1] SYBREN WORM,[1] LUCAS MAY,[1] MARIA VARAKA,[1] THERESA ROHRMOSER,[1] FERNANDO GARCÍA-RAMÍREZ,[1] RENÉ WERKMEISTER,[1] BERNHARD BAUMANN,[1,2] AND CONRAD W. MERKLE[1,*]

[1] *Center for Medical Physics and Biomedical Engineering, Medical University of Vienna, Währinger Gürtel 18-20/4L, 1090 Vienna, Austria*

[2] *Institute of Biomedical Physics, Medical University of Innsbruck, Müllerstraße 44, 6020 Innsbruck, Austria*

[*] *conrad.merkle@meduniwien.ac.at*

Abstract: This template contains important information on submissions to Optica Publishing Group journals. We encourage authors to focus foremost on their content and not on formatting. The template is provided as a guide, but following the visual styling is optional. Each manuscript will be formatted in a consistent way during production. Authors also have the option to submit articles to the Optica Publishing Group preprint server, Optica Open. You may find it helpful to use our optional Paperpal manuscript readiness check and language polishing service. Note that copyright and licensing information should not be added to your journal or Optica Open manuscript.

1. Introduction

Optical coherence tomography angiography (OCTA) is an extension of OCT that visualizes vascular perfusion using the motion contrast from red blood cells (RBCs). Over the years, OCTA has demonstrated widespread acceptance and use for clinical evaluations of vascular features in relation to disease [1]. OCT angiograms can be used to evaluate characteristics on the scale of vascular layers by measuring vascular density or non-perfusion zones, or it can be used to examine individual vessels with measurements of diameter or tortuosity. Nevertheless, OCT is prone to certain artefacts which specifically affect visualization of the vasculature. These artefacts may lead to underestimations of measurements of vessel diameter and density. When viewed in cross-section, large vessels often demonstrate an hourglass appearance with stronger scattering in the center than at the sides (Figure 1A). This appearance is due to orientation effects of RBCs, which align with the blood flow, causing a reduced scattering cross-section at the sides of the vessels. To compensate for these artefacts, OCT contrast agents such as Intralipid have previously been used to increase intravascular intensity signals, particularly in regions where RBC scattering is diminished. Here we examine different methods of measuring vessel size from in vivo OCT scans of mouse retinal vasculature, and we use a contrast agent to obtain ground truth vessel diameter measurements for comparison. We finally propose ways of mitigating or correcting for underestimations of vessel diameter related to typical OCT artefacts.

Optical coherence tomography (OCT) is a non-invasive optical imaging method, which uses interferometry to measure the transit time of light that is backscattered from a sample. OCT has found widespread clinical use in ophthalmology with XXX number of systems and XXX OCT scans performed annually around the world. OCT angiography (OCTA) is a functional extension of OCT which highlights motion between repeated OCT scans to generate images of vascular networks. Typically, OCTA signals are first calculated from the OCT intensity and/or phase information, segmented in 3D manually, algorithmically, or using a deep learning approach, and finally compressed into a 2D image using a maximum intensity projection. These 2D vascular maps can then be binarized by using a global or adaptive threshold, and finally vascular metrics

such as vessel density and diameter can be computed. Of these processing stages, the thresholding step is perhaps the most important. There are dozens of different thresholding methods that have been proposed over the years and each of these may produce different results in the binary map, which would in turn influence the quantification of vascular metrics. In this work, we examine an alternate approach for extracting quantitative vessel diameter by fitting a model to the OCTA data before binarization to avoid thresholding-based effects.

In addition to thresholding-based variability, we hypothesize that OCTA metrics are also affected by OCT artefacts caused by the non-uniform backscattering profiles of red blood cells (RBCs). Because RBCs have a distinct bi-concave disc shape, they more effectively backscatter light when aligned perpendicular to the beam, rather than parallel to it. It has already been observed that in large vessels, a so-called hourglass artefact appears when viewed cross-sectionally. This is cause by the alignment of the RBCs with the vessel walls while under laminar flow, where the RBCs are perpendicular to the beam in the center of the vessels and parallel to the beam a the edges of the vessels. This artefact artificially suppresses the OCT and OCTA signals at the edges of the vessels, making them appear more narrow when projected into an en-face view, which is the preferred way of displaying angiographic data. Here we use contrast enhanced OCT (CE-OCT) to investigate how significant these effects are and how they can be corrected.

2. Methods and Materials

3. Results

3.1. Repeatability and Linearity

-For BW methods, max and mean projections show opposite trends repeatability and linearity as a function of threshold. -Mean intensity projection typically shows improved measurements

4. Discussion

4.1. Maximum Intensity or Mean Intensity?

4.2. Future Directions

A future direction of this work to to move toward fully automated segmentation and calculation vessel diameters from enface angiogram maps. While ROIs were manually selected in this work, it is also possible to automatically segment angiographic data using traditional skeletonization methods, which are commonly used for automated OCTA analysis. From a skeletonized image, vessel angle can be computed for further extraction of vascular cross-sections perpendicular to the vessel axis. These cross-sections could then be processed using the model-based methods described here to enable fully automated analysis of 3D OCTA data sets, rather than the semi-automated approach shown here for 2D OCTA data.

4.3. Computation Time

It is clear from the results that model-based approaches outperform traditional diameter measurements based on a binarized image. Nevertheless, the improved linearity and repeatability come at the cost of computation time. Fitting a model to the raw aniogram data is significantly more computationally expensive and may not always be the most appropriate choice if results of the processing are time sensitive. Nevertheless, these approaches are highly parallelizable given the independent nature of the measurements, so it is expected that compuation time could be significantly accelerated with appropriate hardware.

4.4. Threshholding

The results presented here used a basic, global threshold for the binarization of the angiogram data. There are numerous other approaches for binarizing the angiogram, which will most likely

each behave differently. For example, angiogram signals can be enhanced with vesselness filters and dynamic thresholds can be used to binarize images with varying SNR. When applying a vesselness filter, it is possible to enhance the shape of the angiogram and omit noise, but it is also possible for such enhancement to overcompensate the signal, leading to larger than expected vessel appearance. Dynamic thresholding can also yield a more uniform angiogram appearance, but similarly risks over- or undercompensating vessels given that threshold level directly affects the measured diameter. Further analysis of these other methods would be required to determine their efficacy in returning accurate and precise diameter values.

5. Corresponding author

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6. Abstract

The abstract should be limited to approximately 100 words. If the work of another author is cited in the abstract, that citation is written out as, for example, Opt. Express **32**, 32643 (2024), and a separate citation should be included in the body of the text. The first reference cited in the main text must be [1].There is no need to include numbers, bullets, or lists inside the abstract. Do not add the licensing or copyright statement at submission.

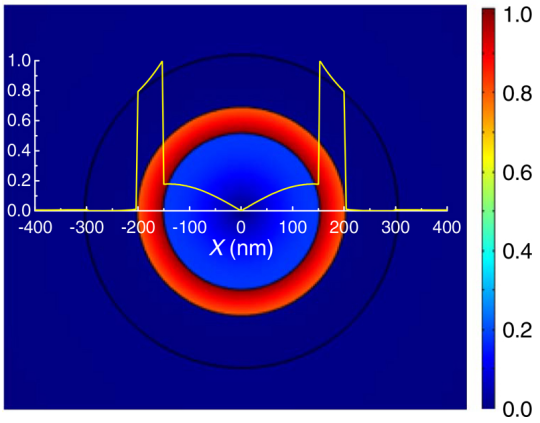

Fig. 1. Sample caption (Fig. 2, [2]).

7. Assessing final manuscript length

The Universal Manuscript Template is based on the Express journal layout and will provide an accurate length estimate for *Optics Express*, *Biomedical Optics Express*, *Optical Materials Express*, *Optica Quantum*, and *Optics Continuum*. *Applied Optics*, JOSAA, JOSAB, *Optics Letters*, *Optica*, *Optica Quantum*, *Optics Continuum*, and *Photonics Research* publish articles in a two-column layout. To estimate the final page count in a two-column layout, multiply the manuscript page count (in increments of 1/4 page) by 60%. For example, 11.5 pages in the Universal Manuscript Template are roughly equivalent to 7 composed two-column pages. Note that the estimate is only an approximation, as treatment of figure sizing, equation display, and other aspects can vary greatly across manuscripts. Authors of Letters may use the length-check template for a more-accurate length estimate.

8. Figures, tables, and supplementary materials

8.1. Figures and tables

Figures and tables should be placed in the body of the manuscript. Standard LaTeX environments should be used to place tables and figures:

```
\begin{figure}[htbp]
\centering\includegraphics[width=7cm]{opticafig1}
\caption{Sample caption (Fig. 2, \cite{Yelin:03}).}
\end{figure}
```

For figures that are being reprinted by permission of the publisher, include the permission line required by that publisher in the caption. Please refer to the information on permissions and reprints for guidance on figure permissions and image use.

To meet accessibility requirements, do not rely solely on color to identify elements in figures (such as blue and red curves). Instead, use shapes or other features along with color. For example, you can use dashed and dotted lines, different shapes for data points, text labels pointing to the color features, numbering, etc.

Table 1, below, provides a formatting example. Note that tables should have titles and not captions. To include additional information, use footnotes as shown in this example:

Table 1. Shape Functions for Quadratic Line Elements

Local Node	$\{N\}_m$	$\{\Phi_i\}_m$ $(i = x, y, z)$
$m = 1$	$L_1(2L_1 - 1)$	Φ_{i1}
$m = 2$	$L_2(2L_2 - 1)$	Φ_{i2}
$m = 3$	$L_3 = 4L_1L_2$	Φ_{i3}

See Table 2 for examples of how to format different types of table cells.

Table 2. Examples of Table Markup[a,b]

<table>
<tr><th colspan="5">Ten normal cells</th></tr>
<tr><td>cell1</td><td>cell2</td><td>cell3</td><td>cell4</td><td>cell5</td></tr>
<tr><td>cell6</td><td>cell7</td><td>cell8</td><td>cell9</td><td>cell10</td></tr>
<tr><th colspan="5">cell1 and cell2 merged.</th></tr>
<tr><td colspan="2">cell1 and cell2</td><td>cell3</td><td>cell4</td><td>cell5</td></tr>
<tr><td>cell6</td><td>cell7</td><td>cell8</td><td>cell9</td><td>cell10</td></tr>
<tr><th colspan="5">cell1 and cell6 merged</th></tr>
<tr><td rowspan="2">cell1
cell6</td><td>cell2</td><td>cell3</td><td>cell4</td><td>cell5</td></tr>
<tr><td>cell7</td><td>cell8</td><td>cell9</td><td>cell10</td></tr>
<tr><th colspan="5">A mix of the two over twenty cells</th></tr>
<tr><td colspan="2">cell1 and cell2</td><td>cell3</td><td>cell4</td><td>cell5</td></tr>
<tr><td rowspan="3">cell6
cell11
cell16</td><td>cell7</td><td>cell8</td><td>cell9</td><td>cell10</td></tr>
<tr><td>cell12</td><td>cell13</td><td>cell14</td><td>cell15</td></tr>
<tr><td>cell17</td><td>cell18</td><td>cell19</td><td>cell20</td></tr>
<tr><th colspan="5">Math environments in merged cells</th></tr>
<tr><td>cell1</td><td>cell2</td><td>cell3</td><td rowspan="2">$\frac{-b\pm\sqrt{b^2-4ac}}{2a}$</td></tr>
<tr><td>cell6</td><td>cell7</td><td>cell8</td></tr>
</table>

[a]Please see the source of this file for macros used.
[b]You can find further table support at Overleaf.

8.2. Supplementary materials in Optica Publishing Group journals

Our journals allow authors to include supplementary materials as integral parts of a manuscript. Such materials are subject to peer-review procedures along with the rest of the paper and should be uploaded and described using our Prism manuscript system. Please refer to the Author Guidelines for Supplementary Materials in Optica Publishing Group Journals for more information on labeling supplementary materials and your manuscript. For preprints submitted to Optica Open a link to supplemental material should be included in the submission.

Authors may also include Supplemental Documents (PDF documents with expanded descriptions or methods) with the primary manuscript. At this time, supplemental PDF files are not accepted for partner titles, JOCN and *Photonics Research*.

8.3. Sample Dataset Citation

1. M. Partridge, "Spectra evolution during coating," figshare (2014), http://doi.org/10.6084/m9.figshare.1004612.

8.4. Sample Code Citation

1. C. Rivers, "Epipy: Python tools for epidemiology," figshare (2014), http://doi.org/10.6084/m9.figshare.1005064.

9. Mathematical and scientific notation

9.1. Displayed equations

Displayed equations should be centered. Equation numbers should appear at the right-hand margin, in parentheses:

$$J(\rho) = \frac{\gamma^2}{2} \sum_{k\,(\mathrm{even})=-\infty}^{\infty} \frac{(1 + k\tau)}{\left[(1 + k\tau)^2 + (\gamma\rho)^2\right]^{3/2}}. \tag{1}$$

All equations should be numbered in the order in which they appear and should be referenced from within the main text as Eq. (1), Eq. (2), and so on [or as inequality (1), etc., as appropriate].

10. Back matter

Back matter sections should be listed in the order Funding/Acknowledgment/Disclosures/Data Availability Statement/Supplemental Document section. An example of back matter with each of these sections included is shown below. The section titles should not follow the numbering scheme of the body of the paper.

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   Data availability. Data underlying the results presented in this paper are available in Dataset 1, Ref. [8].

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   Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

4. If no data were generated or analyzed in the presented research, that should be stated.

   Data availability. No data were generated or analyzed in the presented research.

Data availability statements are not required for preprint submissions.

Supplemental document. A supplemental document must be called out in the back matter so that a link can be included. For example, "See Supplement 1 for supporting content." Note that the Supplemental Document must also have a callout in the body of the paper.

11. References

Proper formatting of references is important, not only for consistent appearance but also for accurate electronic tagging. Please follow the guidelines provided below on formatting, callouts, and use of BibTEX.

11.1. Formatting reference items

Each source must have its own reference number. Footnotes (notes at the bottom of text pages) are not used in our journals. List up to three authors, and if there are more than three use *et al.* after that. Examples of common reference types can be found in the Author Style Guide.

The commands `\begin{thebibliography}{}` and `\end{thebibliography}` format the section according to standard style, showing the title
References. Use the `\bibitem{label}` command to start each reference.

11.2. Formatting reference citations

References should be numbered consecutively in the order in which they are referenced in the body of the paper. Set reference callouts with standard `\cite{}` command or set manually inside square brackets [1].

To reference multiple articles at once, simply use a comma to separate the reference labels, e.g. `\cite{Yelin:03,Masajada:13,Zhang:14}`, produces [2–4].

11.3. BibTEX

BibTEX may be used to create a file containing the references, whose contents (i.e., contents of `.bbl` file) can then be pasted into the bibliography section of the `.tex` file. A BibTEX style file, `opticajnl.bst`, is provided.

If your manuscript already contains a manually formatted `\begin{thebibliography}`... `\end{thebibliography}` list, then delete the `latexmkrc` file (if present) from your submission files.

12. Conclusion

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Add references with BibTeX or manually [2–9].

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9. C. Rivers, "Epipy: Python tools for epidemiology," figshare (2014) [retrieved 13 May 2015], http://dx.doi.org/10.6084/m9.figshare.1005064.